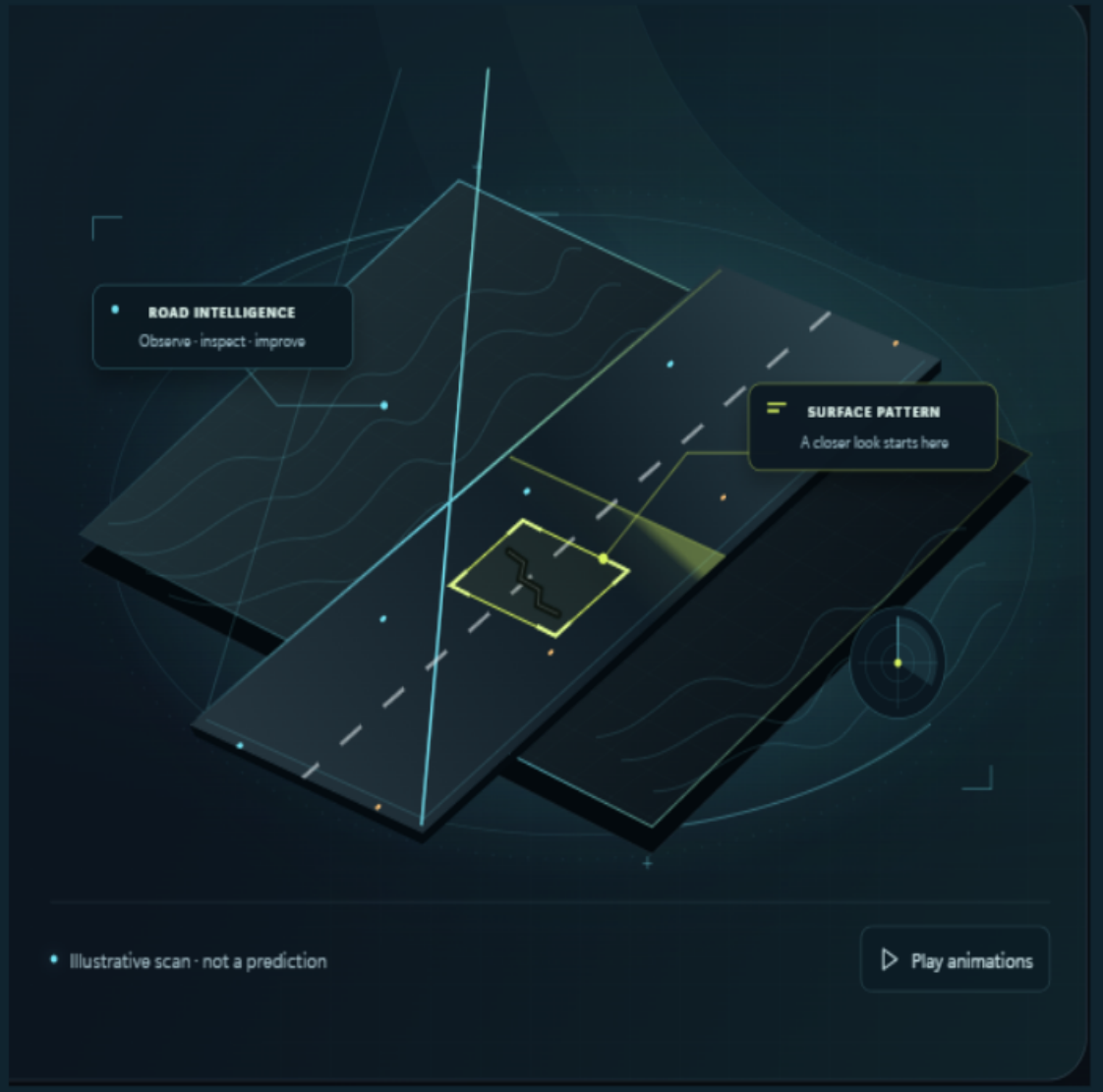
RoadScan

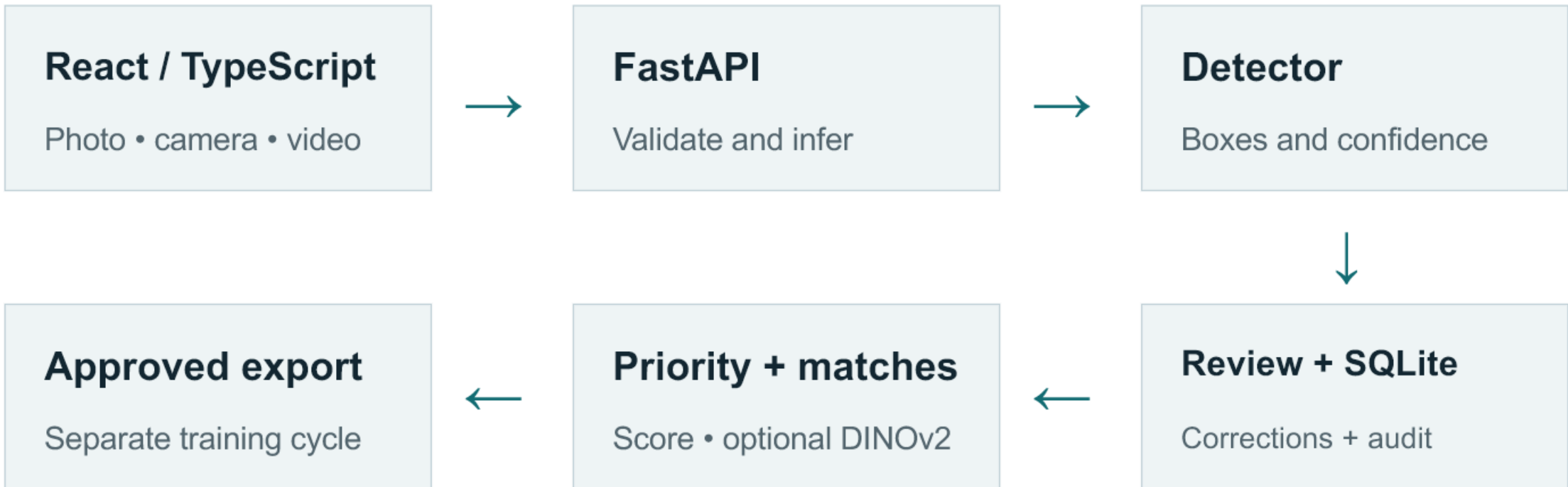
Road damage recognition with human review

Locate visible damage.
Review the evidence.
Prioritize follow-up.



Feras Alyamani · Ahmed Salamah
Ahmed Alhejaili

A local inspection workflow



FastAPI runs inference locally; SQLite and private files retain reports, model files and review history.

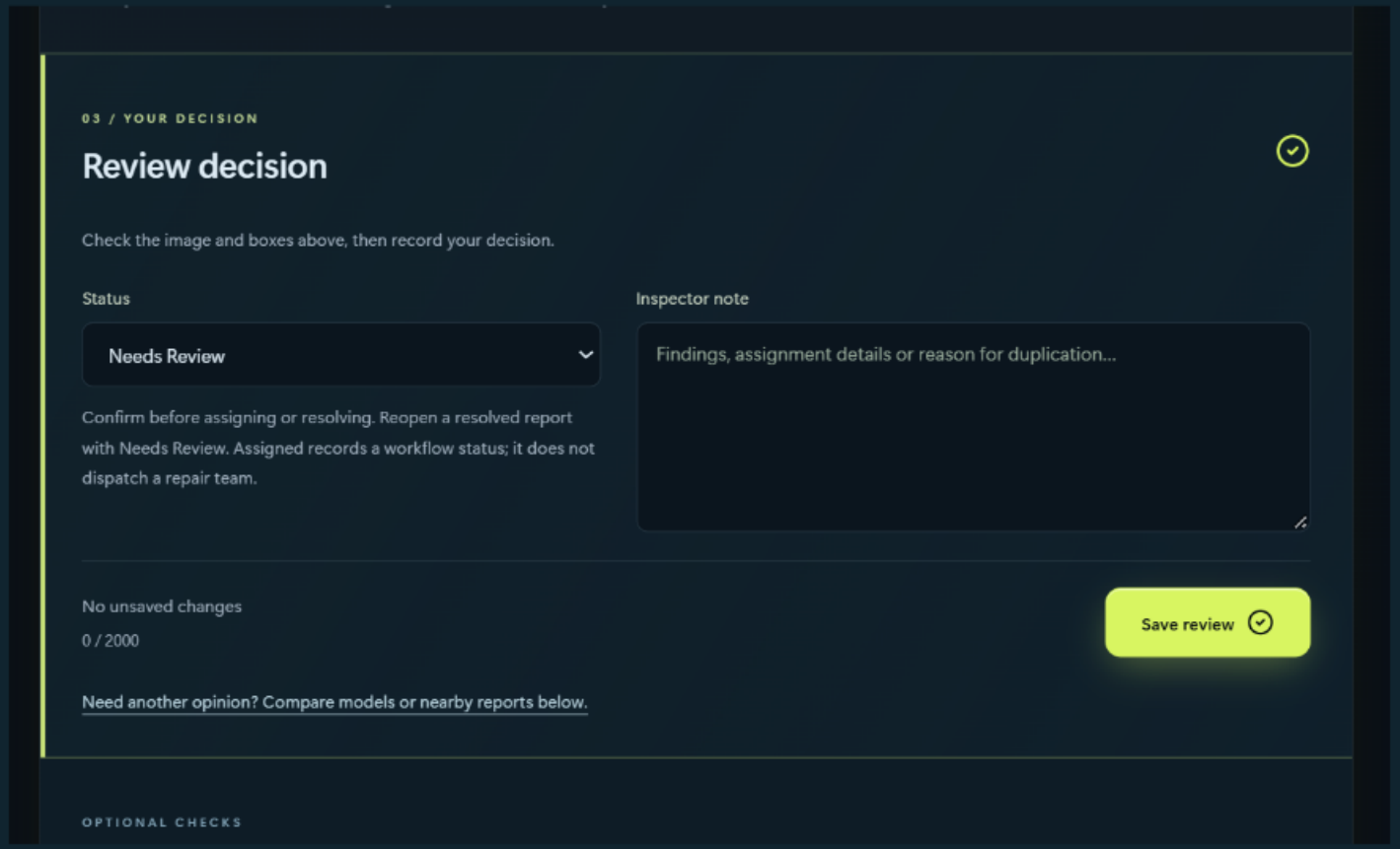
Every prediction stays reviewable

Inspect the boxes

Correct locations or labels.
Save a note and review status.

Keep the human decision

Model voting is advice.
A reviewer can choose a single model even against a majority.



Public edition: owned review controls replace a road-photo example; workflow illustration, not accuracy evidence.

Model evidence and its limits



YOLO26s · Main · 1024 px

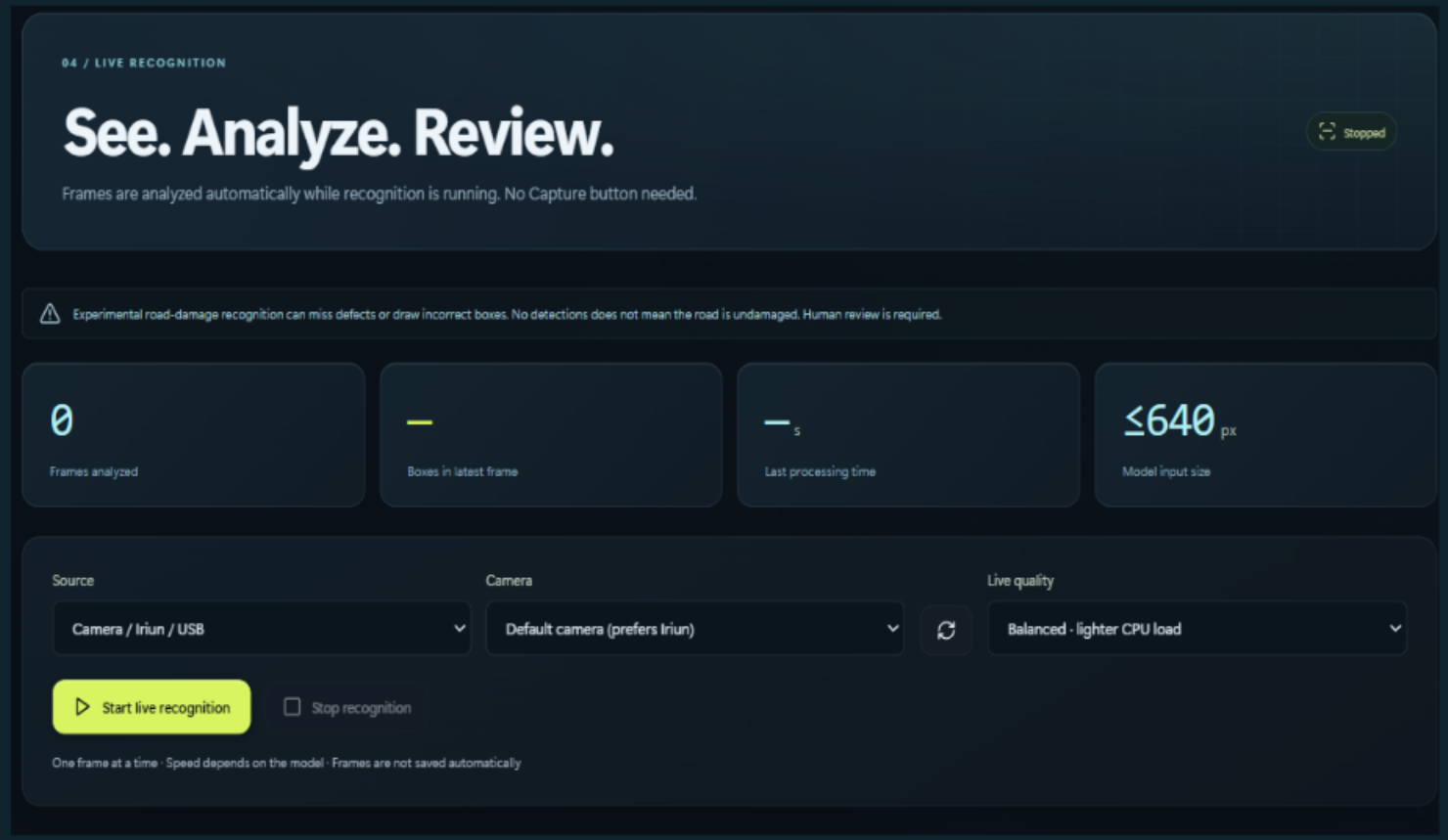
63.04% mAP50
33.01% mAP50–95

Team-reported RDD holdout
Independent reproduction pending
Split leakage may inflate scores

mAP evaluates boxes and labels; 63.04% is not the percentage of road images judged correctly.

AP50 by class: D00 longitudinal · D10 transverse · D20 alligator · D40 pothole.

Camera preview and sequential video analysis



Camera / Iriun / USB

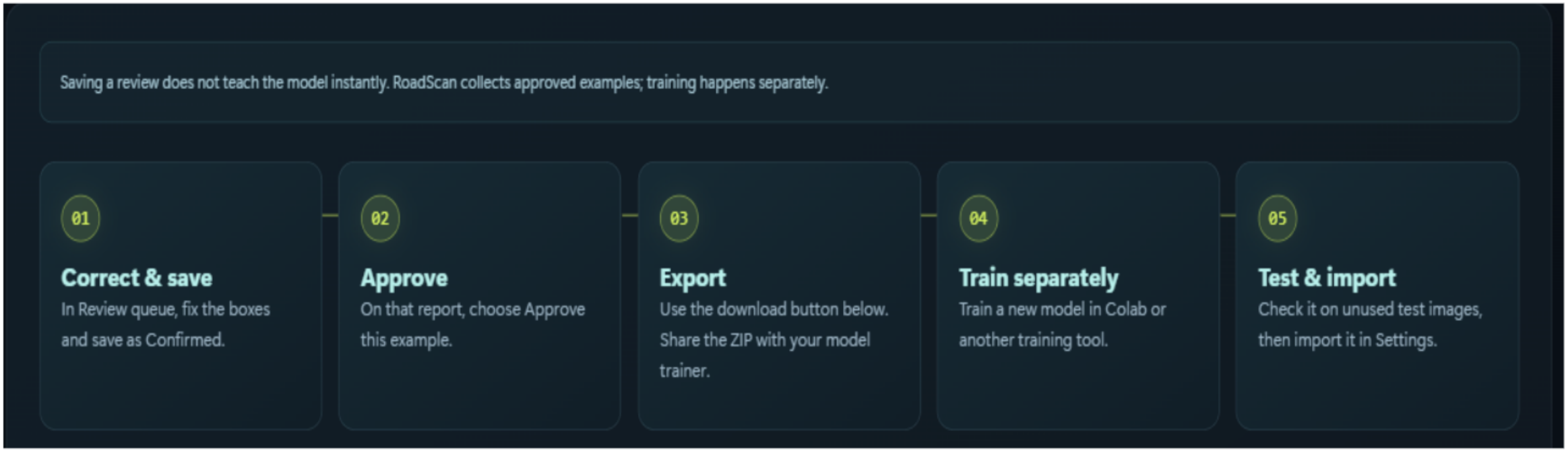
Samples frames automatically, one inference at a time. Lighter models reduce delay.

Video file

The complete-frame job processes decoded frames and may run slower than playback.

Current camera setup screen, 27 Oct · No camera latency or drone-field accuracy is claimed.

Human review prepares the next training cycle



A saved correction does not update model weights.

Approve → export → train separately → evaluate → import the new model.

The current Learning screen makes the approval boundary visible; the example has no approved training rows yet.

Software verified; field validation next

221
backend tests

24
frontend tests

67
browser checks

These checks verify software behaviour, not detection accuracy.

Next evidence

Labeled Saudi road images · real repeat-damage pairs
Repaired data splits · representative camera latency

Local Docker release and setup guide ready · 1 optional backend test skipped · Demo examples remain separate.

Appendix · Data and metric definitions

26,869

reported RDD training images

≈41,000

reported main-run pool with extra sources

5,758 validation + 5,758 labeled holdout images

Reported leakage: 2.2% validation / 2.4% holdout.
Raw split manifests and source licenses remain to be checked.

mAP50: box-and-class precision across recall at IoU 0.50.
mAP50–95: averages stricter overlap thresholds; image-presence F1 is a different task.

Team-reported counts, not independently counted; extra sources and ablations do not share one universal training total.

Appendix · Baseline and model size

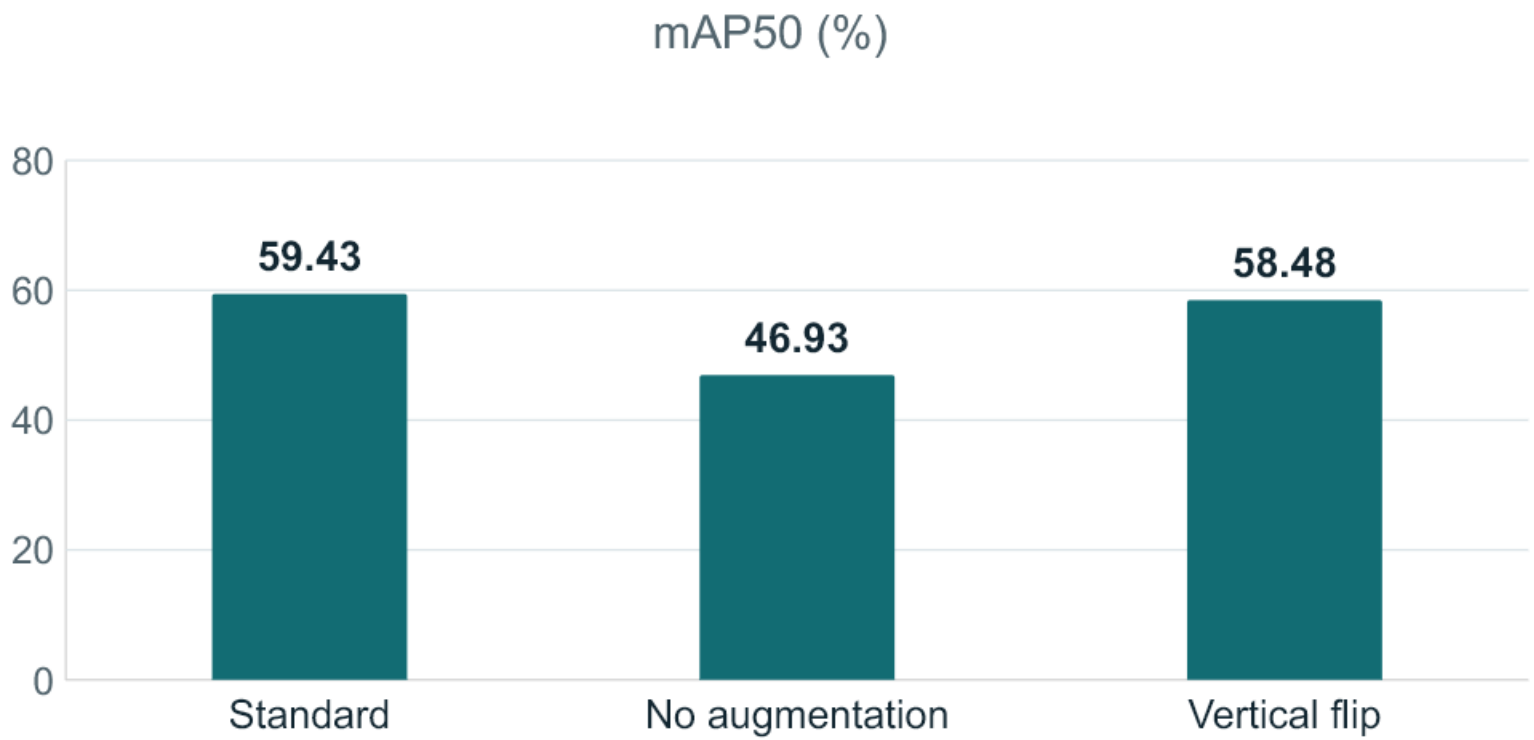
E1 image presence	Any damage F1	Image accuracy
ResNet18 baseline	0.893	0.869
Main YOLO26s	0.929	0.913

E2 Small: 59.43% mAP50 Nano: 55.99%

Nano was faster in a limited local synthetic diagnostic; measure end-to-end latency on the real camera.

Supplied labeled-holdout results; independently reproduced road accuracy remains pending

Appendix · Augmentation and domain shift



India holdout

8.86%

mAP50 with India excluded from comparison training

39.25% when India is included in comparison training

Teammate-reported experiments; different evaluation domains must remain separate

Appendix · Questions and demo fallback

Is priority a severity measurement?

No: a 0–100 heuristic uses type 40%, visible area 25%, count 20% and confidence/review 15%; it does not measure physical depth.

Can duplicate reports be merged safely?

Optional DINOv2 + FAISS searches nearby candidates within 50 m. The 0.80 threshold is experimental; reports are never merged automatically.

What if the live demonstration fails?

Use the PDF screenshots and describe the saved workflow. Restart Docker; check the local URL and active model. Do not claim a failed run succeeded.

How do we reproduce the project?

Follow README → Docker setup → token → import a trusted model → Use model. Source, weights and private data are distributed separately.

References: Team-Models-and-Results.md · Replacement-Tools.md · README.md · For-Examiners.md